

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Date:	03-08-2026	Duration:	60 minutes

Code of Conduct

I A.Akshaya(192425170) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Akshaya

ACADEMIC PROJECT, ALLOCATION MONITORING AND EVALUATION MANAGEMENT SYSTEM

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. Study and Analyze the Scenario

- Carefully understand the given problem statement.
- The current project allocation system mainly depends on students' **CGPA or academic performance**.
- Students' **interests, talents, and technical skills** are often not considered during project allocation.
- Every student has different strengths and preferred domains, such as AI, Web Development, Cyber Security, or Data Science.
- Assigning projects without considering students' interests may reduce their motivation and learning experience.
- Students may face difficulties in completing projects that are outside their area of interest.
- Faculty members spend more time manually monitoring student progress and project submissions.
- The existing system lacks a transparent and organized method for project allocation.
- There is no proper platform to record students' interests and programming skills.
- Tracking project progress and maintaining evaluation records manually can be time-consuming.
- Communication between students, faculty, and project coordinators may become inefficient.
- The proposed system aims to automate project allocation, monitoring, and evaluation.
- Projects will be assigned based on students' **interests, technical skills, and preferred domains**.
- Faculty members can easily monitor progress and provide continuous guidance.
- Management and HOD can track project status and generate reports efficiently.

- The overall goal is to create a **fair, transparent, and student-centered project management system** that improves learning outcomes and career opportunities.
- Identify the business domain, stakeholders, existing challenges, and system objectives.

Business Domain

The project belongs to the **Education** domain. It is mainly designed for colleges and universities to manage the process of allocating, monitoring, and evaluating students' academic projects.

Stakeholders

The people involved in this system are:

- **Students** – Complete the assigned projects and submit their work.
- **Faculty Guides** – Guide students and evaluate their project progress.
- **Project Coordinator** – Assign projects, monitor deadlines, and manage the project process
- **HOD (Head of Department)** – Supervise and approve project-related activities.
- **College Management** – Monitor overall project performance and view reports.
- **System Administrator** – Manage user accounts, project details, and maintain the system.
- **External Examiner** – Conduct the final project evaluation and award marks.

Existing Challenges

- Projects are mostly assigned based on **CGPA** instead of students' interests.
- Students may not get projects in their preferred domain.
- Manual project allocation and monitoring take more time.
- It is difficult to track the progress of every student.

System Objectives

The main objective of the **Academic Project Allocation, Monitoring and Evaluation Management System** is to allocate projects based on students' interests, technical skills, and preferred domains instead of considering only their CGPA. This helps students work on projects they enjoy and understand better, which improves their learning experience. The system also aims to enhance students' practical knowledge by giving them opportunities to apply their skills

in real-world projects. Another important objective is to make the project allocation process fair, transparent, and free from bias. The system allows faculty members to monitor students' project progress regularly, provide guidance, and evaluate their performance effectively. It maintains all project-related information, including student profiles, project details, progress reports, and evaluation records, in a single centralized system, making it easy to access and manage. The system also improves communication between students, faculty members, project coordinators, HOD, and management by providing a common platform for sharing updates and feedback.

2. Identify Functional and Non-Functional Requirements

- List all functional requirements required by the proposed system.

Functional Requirements

- Student registration and secure login.
 - Faculty, HOD, Project Coordinator, Admin, and Examiner login.
 - Student profile creation and profile update.
 - Students can select their **area of interest**, preferred domain, and programming skills.
 - Students can view the projects allocated to them.
 - Project Coordinator can allocate projects based on students' interests and technical skills.
 - Faculty members can view assigned students and monitor their project progress.
 - Students can upload project reports, documents, and weekly progress updates.
 - Faculty can review submissions, provide feedback, and assign internal marks.
 - External Examiner can evaluate the final project and submit marks.
 - HOD and Management can monitor overall project progress.
 - Generate project status reports and evaluation reports.
 - Send notifications for project reviews, deadlines, and submissions.
 - Maintain a centralized database for students, faculty, projects, and evaluations.
 - Admin can manage users, projects, departments, and system settings.
- Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.

The non-functional requirements are:

- **Performance:** The system should load pages quickly and respond to user requests within a few seconds, even when many users are accessing it at the same time.

- **Security:** Student records, project details, marks, and evaluation reports should be protected using secure login, passwords, and role-based access control.
- **Reliability:** The system should work correctly without frequent failures and should store project data accurately without losing information.
- **Usability:** The application should have a simple, user-friendly interface so that students, faculty, HOD, and administrators can use it easily without special training.
- **Scalability:** The system should be able to support an increasing number of students, faculty members, departments, and projects without affecting its performance.
- **Maintainability:** The software should be easy to update, fix, and improve whenever new features or changes are required.
- **Availability:** The system should be available whenever users need it, with minimal downtime, especially during project submission, monitoring, and evaluation periods.
- **Data Integrity:** The system should ensure that all project records, marks, and student information remain accurate, consistent, and free from unauthorized modifications.
- **Backup and Recovery:** Regular backups should be maintained so that project data can be restored in case of system failure or accidental data loss.
- **Compatibility:** The application should work properly on different web browsers and devices such as desktops, laptops, tablets, and smartphones.

3. Identify Actors and Use Cases

- Identify all primary and secondary actors interacting with the system.

Primary Actors

These are the users who directly interact with the system.

- **Student** – Registers, updates profile, selects interests, views allocated project, uploads reports, and checks evaluation results.
- **Faculty Guide** – Monitors student progress, provides guidance, reviews submissions, and evaluates projects.
- **Project Coordinator** – Allocates projects, assigns faculty guides, manages deadlines, and tracks project progress.
- **Head of Department (HOD)** – Monitors project activities, approves project allocation, and reviews reports.

- **System Administrator (Admin)** – Manages users, departments, projects, and maintains the system.
- **External Examiner** – Evaluates final projects and submits final marks.

Secondary Actors

These actors support the system indirectly.

- **College Management** – Views project reports and monitors overall project performance.
- **Database** – Stores student, faculty, project, and evaluation data.
- **Notification Service** – Sends reminders about deadlines, meetings, and project reviews.
 - List the major use cases associated with each actor.

Student	Faculty Guide
Register and Login	Login
Update Profile	View Assigned Students
Select Interests and Skills	Monitor Project Progress
View Allocated Project	Review Reports
Upload Project Reports	Give Feedback
View Feedback and Marks	Evaluate Projects

Faculty Guide

• Login
• View Assigned Students
• Monitor Project Progress
• Review Reports
• Give Feedback
• Evaluate Projects

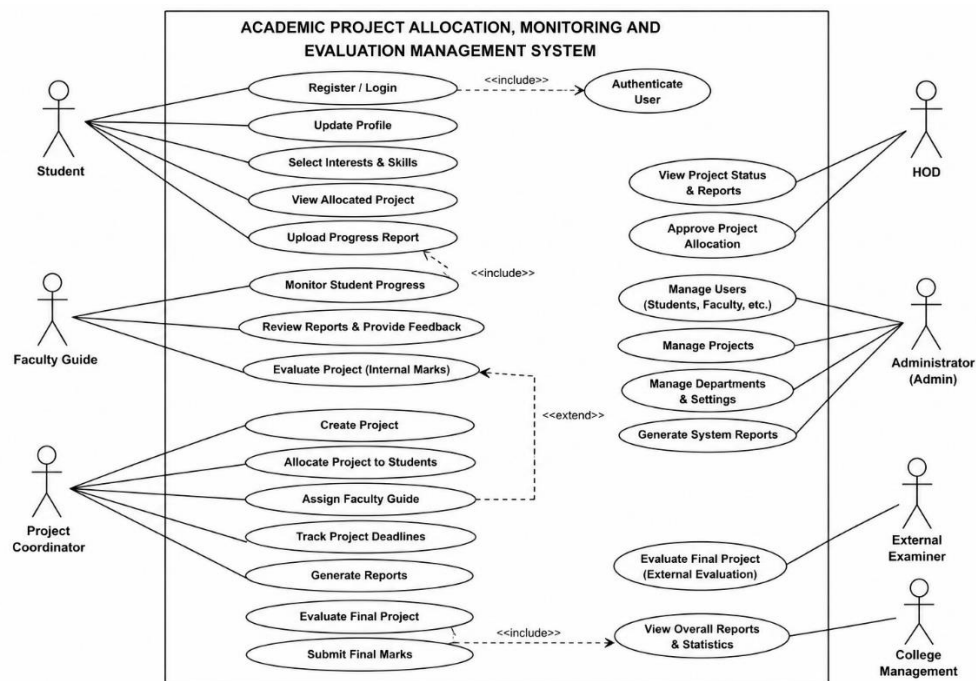
Project Coordinator

• Login
• Create Projects
• Allocate Projects
• Assign Faculty Guides
• Monitor Deadlines
• Generate Reports

HOD	Administrator
Login	Manage Users
View Project Status	Manage Departments
Approve Project Allocation	Manage Projects
Review Reports	Maintain Database & Generate reports

External Examiner	Management
Login	View Overall Reports
View Final Projects	Monitor Student Performance
Evaluate Projects	Review Project Statistics
Submit Final Marks	

- Draw a Use Case Diagram representing the interactions between actors and the system.



4. Prepare the Software Requirement Specification (SRS)

Develop an SRS document containing:

- Introduction

The **Academic Project Allocation, Monitoring and Evaluation Management System** is a web-based application developed to simplify the process of assigning, monitoring, and evaluating academic projects. The system helps students receive projects based on their interests, technical skills, and preferred domains instead of only their CGPA. It also enables faculty members, project coordinators, HODs, and management to monitor project progress and evaluate students efficiently. The main goal is to improve learning, ensure fairness, and reduce manual work in project management.

- Problem Description

In many colleges, academic projects are assigned manually or mainly based on students' academic performance. This approach may ignore students' interests and technical abilities, reducing motivation and learning outcomes. Monitoring project progress and maintaining evaluation records manually is time-consuming and may lead to errors. Therefore, there is a need for an automated system that allocates projects fairly, tracks progress, and simplifies evaluation.

- Scope of the System

- ✓ Register students and faculty members.
- ✓ Store student profiles, interests, and technical skills.
- ✓ Allocate projects based on student preferences.
- ✓ Monitor project progress.
- ✓ Evaluate student performance.
- ✓ Generate reports for management.
- ✓ Maintain project records securely.

- Functional Requirements

User Registration and Login, Student Profile Management, Interest and Technical Skill Selection, Project Allocation based on student interests, Faculty Guide Assignment, Student Dashboard, Faculty Dashboard, Project Progress Upload, Project Monitoring, Internal Evaluation , External Evaluation, Report Generation, Notifications for deadlines, User and Project Management, Database Management.

- Non-Functional Requirements

- ❖ **Performance:** Fast response time during login, project allocation, and report generation.
- ❖ **Security:** Role-based login, password protection, and secure data storage.
- ❖ **Reliability:** Accurate project allocation and uninterrupted system operation.
- ❖ **Usability:** Easy-to-use interface for all users.
- ❖ **Scalability:** Support increasing numbers of students and projects.
- ❖ **Maintainability:** Easy to update and maintain.
- ❖ **Availability:** System should be available whenever users need it.
- ❖ **Compatibility:** Works on different browsers and devices.
- ❖ **Backup & Recovery:** Automatic backup of important project data.

- Use Case Descriptions

Register/Login, Update Profile, Select Interests & Skills, View Allocated Project, Upload Progress Report, Monitor Progress, Allocate Project, Assign Faculty Guide, Evaluate Project, Generate Reports, Manage Users, Submit Final Marks, View Dashboard, Logout

- Data Requirements

Student Details (Name, Roll Number, Department, Email), Faculty Details, Student Interests and Skills, Project Details, Project Allocation Information, Weekly Progress Reports, Internal and External Evaluation Marks, Feedback Records, Deadlines, User Login Information, Report Data

- External Interface Requirements

User Interface

- Simple and user-friendly web interface.
- Responsive design for desktop and mobile devices.

Hardware Interface

- Computer/Laptop
- Smartphone
- Internet Connection

Software Interface

- Frontend: HTML, CSS, JavaScript, React (optional)
- Backend: Java / Python / Node.js
- Database: MySQL

Communication Interface

- Internet
 - Email notifications
 - SMS notifications (optional)
- System Features

Secure User Registration and Login, Student Profile Management, Interest-Based Project Allocation, Faculty Guide Assignment, Student and Faculty Dashboards, Project Progress Tracking, Internal and External Evaluation, Feedback Management, Automatic Report Generation, Deadline Notifications, User and Project Management, Secure Database Storage, Search and Filter Options, Performance Monitoring, Role-Based Access Control

5. Identify Assumptions and Constraints

- List the assumptions considered while developing the system.

Allocation, Monitoring and Evaluation Management System:

- All students and faculty members have valid login credentials.
- Students will provide correct personal details, interests, and technical skills.
- Faculty members will regularly monitor and evaluate student projects.
- Project Coordinators will allocate projects fairly based on students' interests and skills.
- HOD and Management will use the system to monitor project progress.
- Users will have access to a computer or mobile device with an internet connection.
- The database will securely store all project-related information.
- The institution has the required hardware and software to run the application.
- All users will follow the guidelines and use the system responsibly.
- Regular backups will be taken to prevent data loss.

- Identify technical, operational, economic, legal, and time-related constraints affecting the project.

Technical Constraints

- Requires a stable internet connection.
- Depends on reliable server and database availability.

Operational Constraints

- Users may require basic training to use the system.
- Faculty and students must regularly update project information.

Economic Constraints

- Development, hosting, and maintenance require financial resources.
- Software licenses, server costs, and database management may increase the project cost.

Legal Constraints

- Student and faculty data must be kept confidential.
- The system must comply with the institution's privacy and data protection policies.

Time-Related Constraints

- The project must be completed within the academic schedule.
- Sprint deadlines should be followed for timely development.

6. Validate Requirement Completeness and Consistency

- Verify whether all stakeholder requirements have been addressed.

The proposed system has been designed by considering the requirements of all the important stakeholders. Students can register, update their profiles, select their interests and technical skills, view allocated projects, upload progress reports, and check evaluation results. Faculty members can monitor student progress, provide guidance, and evaluate projects. The Project Coordinator can allocate projects, assign faculty guides, and manage deadlines. The HOD can supervise project activities and review reports. The Administrator can manage users, projects, and the overall system. College Management can view reports and monitor project performance.

- Check for ambiguity, redundancy, inconsistency, and missing requirements.

The system requirements are written in a clear and understandable manner to avoid ambiguity. Similar requirements are grouped together to eliminate redundancy and prevent duplication. The functional and non-functional requirements are consistent with the project objectives and stakeholder needs. However, a few additional requirements can improve the system, such as email notifications for project deadlines, automatic reminders for faculty reviews, document version control, and a project search feature. These additions will make the system more complete and user-friendly.

- Suggest improvements to enhance the quality and completeness of the requirement specification.

The quality of the requirement specification can be improved by adding more detailed use case descriptions and activity diagrams for each module. Security can be strengthened by implementing two-factor authentication and encrypted password storage. A dashboard with charts and project analytics can help management monitor student performance more effectively. Features such as email and SMS notifications, online feedback forms, automatic backup, cloud storage, and a mobile-friendly interface can improve usability. Regular reviews and updates of the requirements based on stakeholder feedback will ensure that the system continues to meet the needs of students, faculty, and management. These improvements will make the proposed system more efficient, reliable, secure, and suitable for future enhancements.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional	Clearly identifies comprehensive, accurate, and well-	Identifies most requirements with minor	Identifies only basic requirements;	Requirements are incomplete, inaccurate, or

Requirement Identification (25%)	classified functional and non-functional requirements.	classification errors.	several omissions.	poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Mark
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25

